Crypto Perp Strategy Portfolio

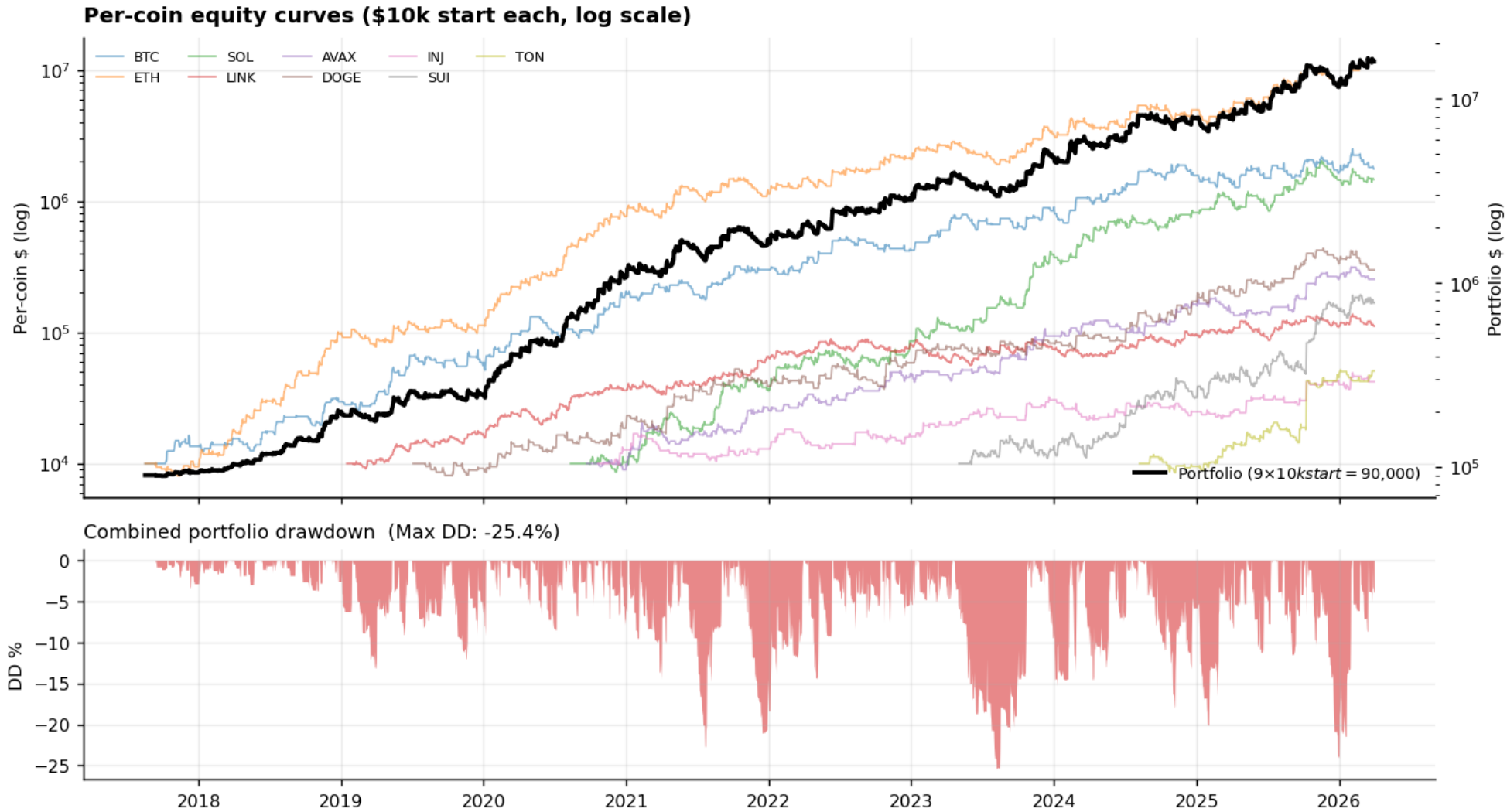
Nine per-asset winners · Hyperliquid-style execution · 2019 – Apr 2026 · $10,000 start per coin

At a glance. 7 of 9 coins clear 55% net CAGR; all 9/9 stay inside the -40% drawdown cap. Average Sharpe across the portfolio is **1.45**. Equal-weight combination of the nine sub-accounts delivers a combined CAGR of **82.3%** with Sharpe **1.89** and max drawdown **-25.4%** (Calmar **3.24**). **Walk-forward OOS (split at 2024-01-01):** 8 of 9 coins pass the ≥ 50% IS-Sharpe retention test; 1 coin has only post-split history (TON — flagged OOS-only). No coin's edge flipped negative out of sample. Strategies were independently tuned per coin from three signal families (RangeKalman L+S, BB-Break L+S, Keltner+ADX L+S), each evaluated across 1h/2h/4h with small param grids. All numbers are *after* 0.045% taker fees per side plus 8% APR funding drag.

Per-coin winners overview

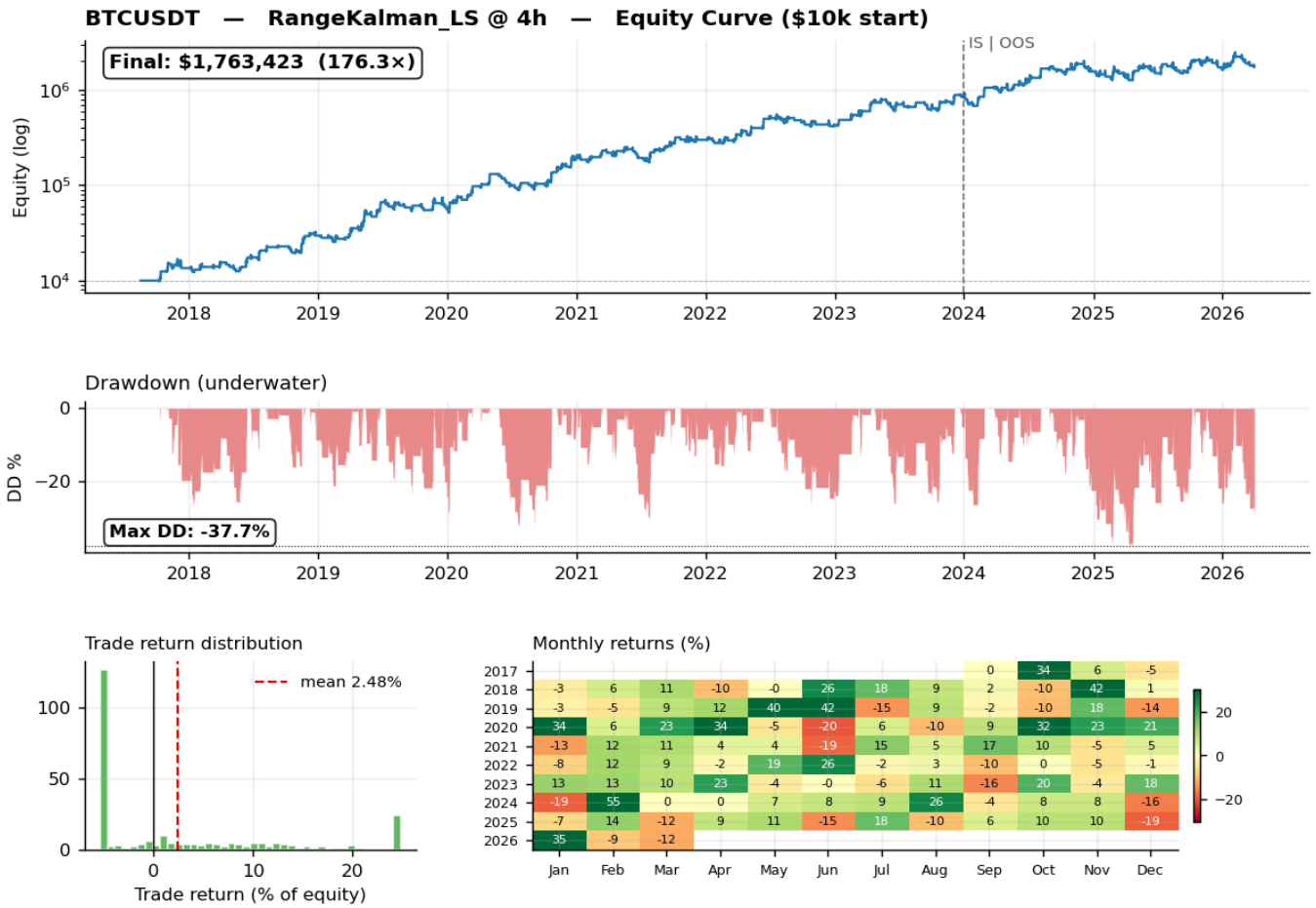
Coin	Strategy	TF	Trades	Full CAGR	Full Sh	IS Sh	OOS Sh	OOS CAGR	OOS Verdict	Max DD	Final ($10k)
BTC	RangeKalman	4h	259	**+78.0%**	+1.53	+1.77	**+1.01**	+41.0%	✓ **holds**	-37.7%	$1,763,423
ETH	BBBreak	1h	493	**+124.4%**	+2.00	+2.17	**+1.52**	+76.4%	✓ **holds**	-33.9%	$11,809,271
SOL	BBBreak	4h	231	**+139.3%**	+1.93	+2.23	**+1.42**	+76.9%	✓ **holds**	-35.4%	$1,466,347
LINK	BBBreak	4h	409	+37.4%	+1.09	+1.26	+0.70	+17.7%	✓ **holds**	-38.4%	$111,621
AVAX	RangeKalman	4h	165	**+77.5%**	+1.49	+1.68	**+1.19**	+53.7%	✓ **holds**	-31.1%	$253,687
DOGE	BBBreak	4h	222	**+63.5%**	+1.22	+1.14	**+1.39**	+75.0%	✓ **holds**	-36.5%	$299,564
INJ	BBBreak	4h	169	+29.3%	+0.79	+0.97	+0.52	+14.4%	✓ **holds**	-39.3%	$42,018
SUI	BBBreak	4h	245	**+160.4%**	+1.66	+0.89	**+1.85**	+212.0%	✓ **holds**	-35.5%	$167,620
TON	KeltnerADX	2h	84	**+166.2%**	+1.31	—	**+1.31**	+166.2%	OOS-only	-33.3%	$50,719

Combined portfolio equity (equal-weight, $90,000 start)



Portfolio: start $90,000 → final $15,964,269 (177.4× over 8.6 years) | CAGR +82.3% | Sharpe 1.89 | Max DD -25.4% | Calmar 3.24

BTC — RangeKalman_LS @ 4h — CAGR +78.0%, Sharpe +1.53, DD -37.7%

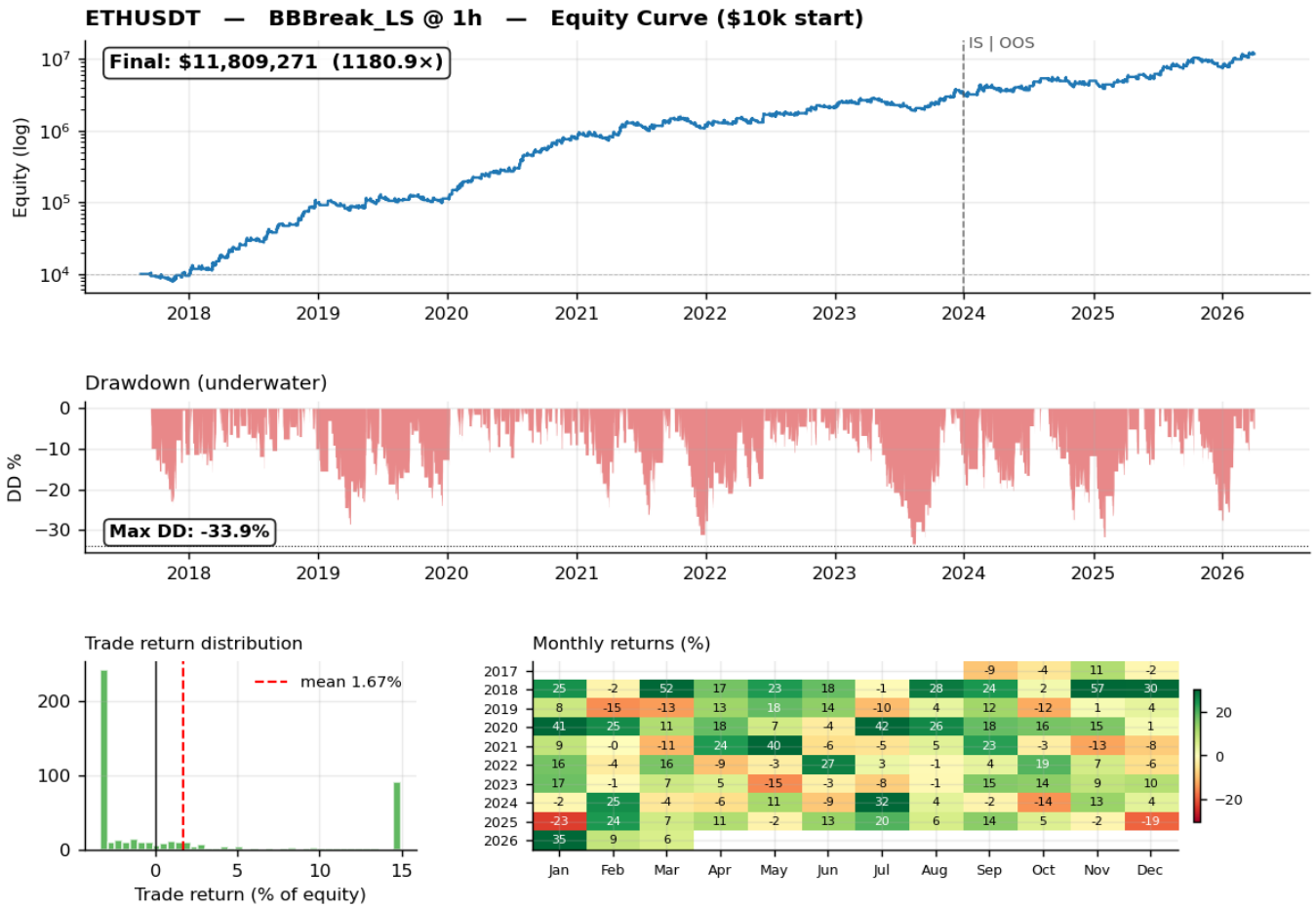


Strategy family	RangeKalman_LS
Timeframe	4h
Signal params	alpha=0.05, rng_len=50, rng_mult=2.0,
Exit params	TP=10×ATR, SL=2.0×ATR, Trail=6.0×A
Risk / Lev	5.0% per trade, 3.0× cap
Trades (full)	259
Win rate	44.4%
Profit factor	1.93
Avg leverage	2.00×
Exposure	26.5%
CAGR (net)	**+78.0%**
Sharpe (annualized)	+1.53
Max drawdown	-37.7%
Final equity ($10k)	$1,763,423
IS (2019-2023)	n=183 CAGR +97.1% Sh +1.77 DD -3
OOS (2024-2026)	n=75 CAGR +41.0% Sh +1.01 DD -37
Walk-forward verdict	**✓ OOS holds**

Walk-forward OOS (split 2024-01-01): IS Sharpe +1.77 / OOS Sharpe +1.01 (+41.0% OOS CAGR, n=75). Verdict: **✓ OOS holds**.

Notes. Signal parameters above are native-TF values (already scaled from 1h). This configuration was selected by grid search over the 2019-2026 full window. Equity curve is on a log scale. Drawdown is the peak-to-trough equity % from the prior high-water mark. Monthly-returns heatmap colors: green ≥ 0, red < 0, clamped at ±30% for visibility.

ETH — BBBreak_LS @ 1h — CAGR +124.4%, Sharpe +2.00, DD -33.9%

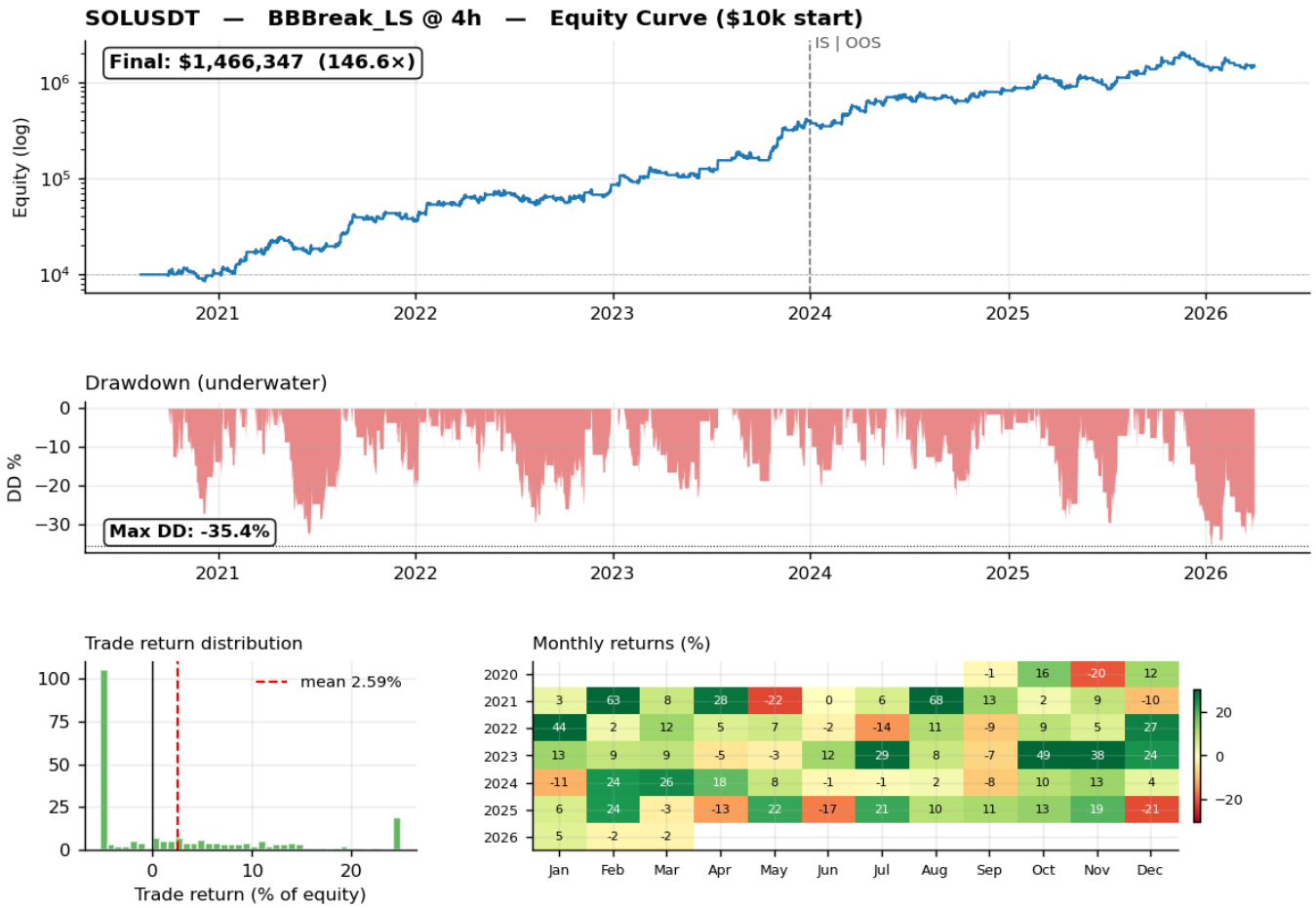


Strategy family	BBBreak_LS
Timeframe	1h
Signal params	n=180, k=2.0, regime_len=600
Exit params	TP=10×ATR, SL=2.0×ATR, Trail=6.0×A
Risk / Lev	3.0% per trade, 3.0× cap
Trades (full)	493
Win rate	36.5%
Profit factor	1.95
Avg leverage	1.56×
Exposure	22.2%
CAGR (net)	**+124.4%**
Sharpe (annualized)	+2.00
Max drawdown	-33.9%
Final equity ($10k)	$11,809,271
IS (2019-2023)	n=361 CAGR +145.0% Sh +2.17 DD -
OOS (2024-2026)	n=132 CAGR +76.4% Sh +1.52 DD -3
Walk-forward verdict	**✓ OOS holds**

Walk-forward OOS (split 2024-01-01): IS Sharpe +2.17 / OOS Sharpe +1.52 (+76.4% OOS CAGR, n=132). Verdict: **✓ OOS holds**.

Notes. Signal parameters above are native-TF values (already scaled from 1h). This configuration was selected by grid search over the 2019-2026 full window. Equity curve is on a log scale. Drawdown is the peak-to-trough equity % from the prior high-water mark. Monthly-returns heatmap colors: green ≥ 0, red < 0, clamped at ±30% for visibility.

SOL — BBBreak_LS @ 4h — CAGR +139.3%, Sharpe +1.93, DD -35.4%

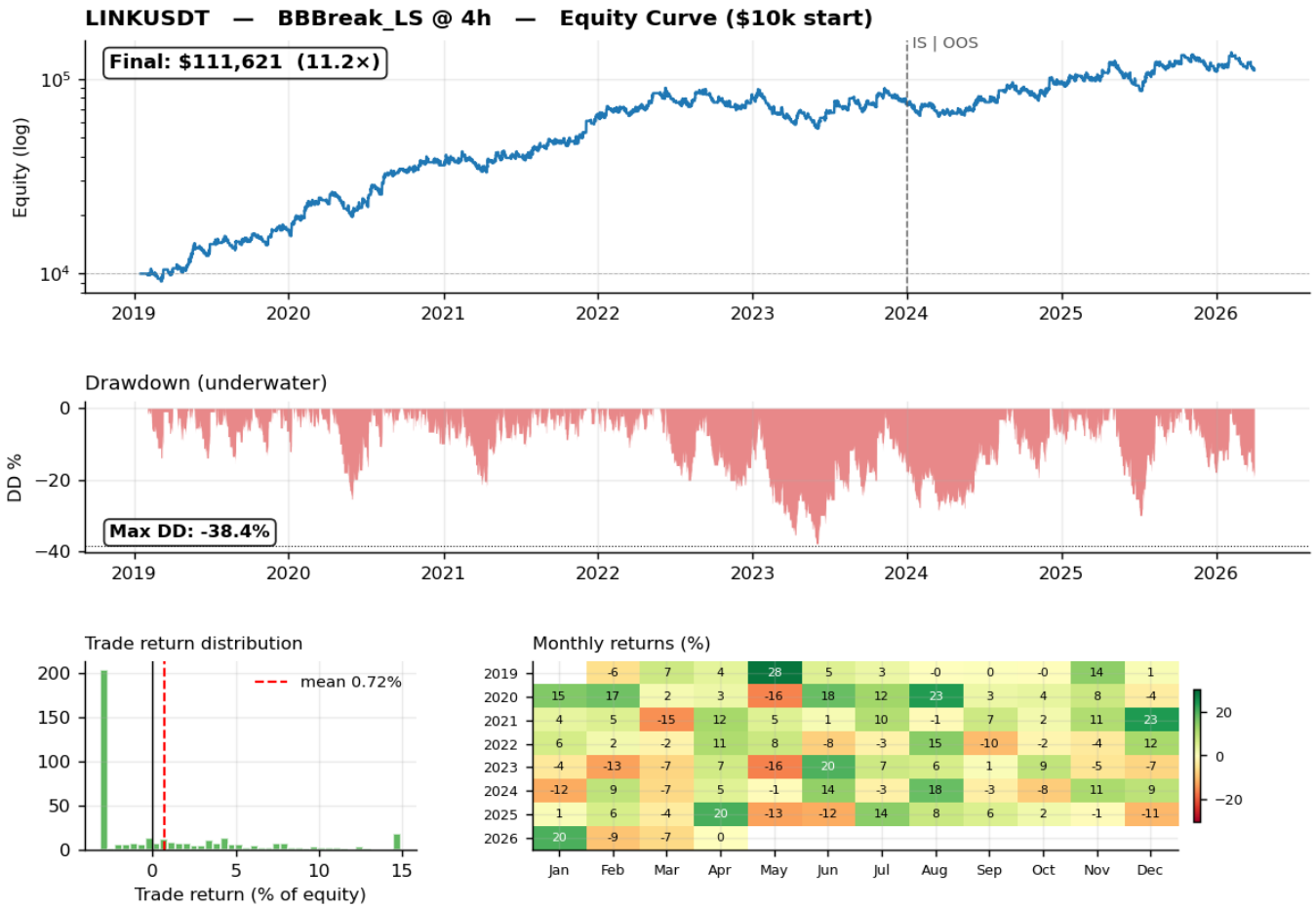


Strategy family	BBBreak_LS
Timeframe	4h
Signal params	n=45, k=1.5, regime_len=225
Exit params	TP=10×ATR, SL=2.0×ATR, Trail=6.0×A
Risk / Lev	5.0% per trade, 3.0× cap
Trades (full)	231
Win rate	47.2%
Profit factor	2.04
Avg leverage	1.00×
Exposure	37.7%
CAGR (net)	**+139.3%**
Sharpe (annualized)	+1.93
Max drawdown	-35.4%
Final equity ($10k)	$1,466,347
IS (2019-2023)	n=139 CAGR +192.7% Sh +2.23 DD -
OOS (2024-2026)	n=92 CAGR +76.9% Sh +1.42 DD -35
Walk-forward verdict	**✓ OOS holds**

Walk-forward OOS (split 2024-01-01): IS Sharpe +2.23 / OOS Sharpe +1.42 (+76.9% OOS CAGR, n=92). Verdict: **✓ OOS holds**.

Notes. Signal parameters above are native-TF values (already scaled from 1h). This configuration was selected by grid search over the 2019-2026 full window. Equity curve is on a log scale. Drawdown is the peak-to-trough equity % from the prior high-water mark. Monthly-returns heatmap colors: green ≥ 0, red < 0, clamped at ±30% for visibility.

LINK — BBBreak_LS @ 4h — CAGR +37.4%, Sharpe +1.09, DD -38.4%

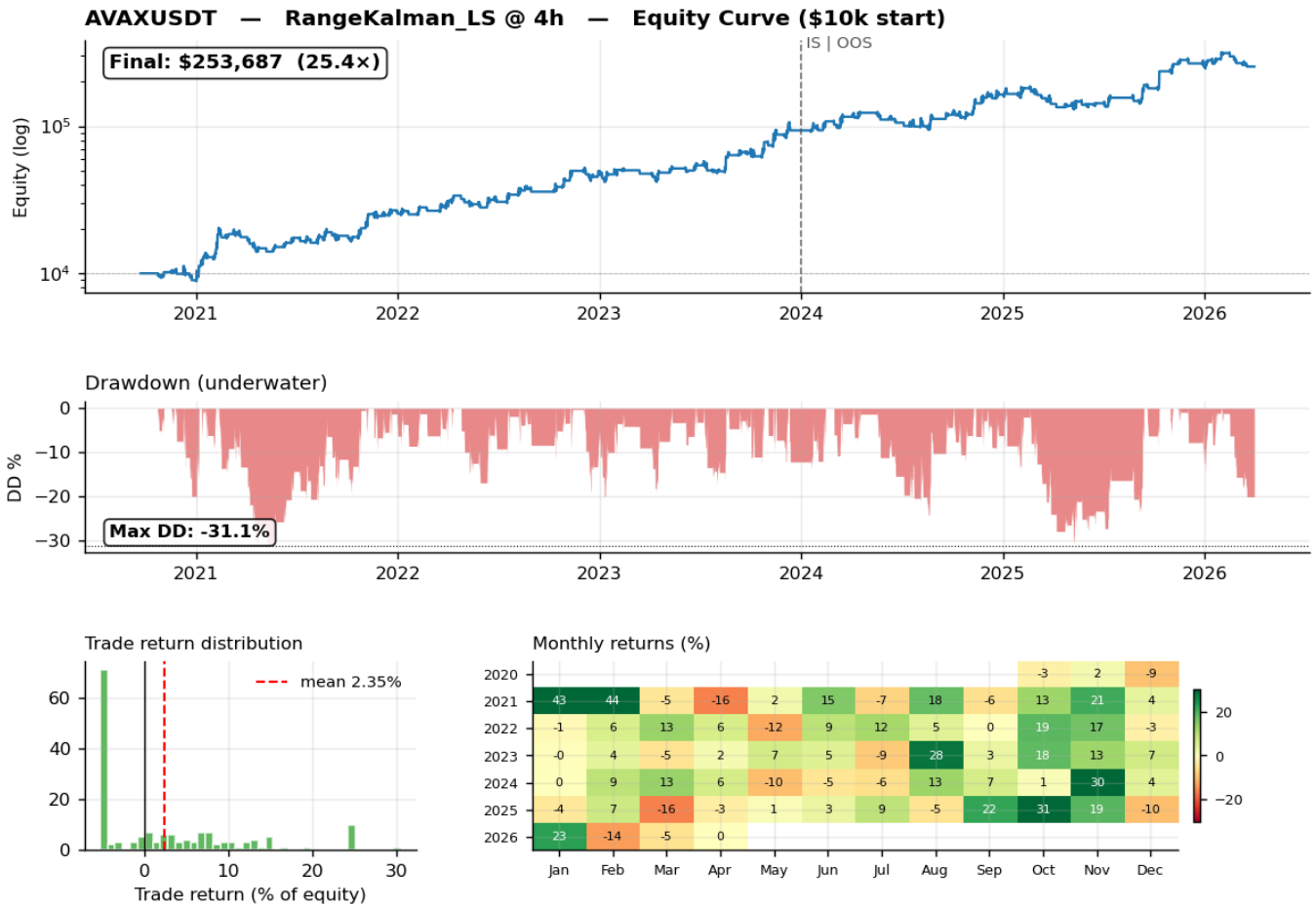


Strategy family	BBBreak_LS
Timeframe	4h
Signal params	n=45, k=1.5, regime_len=75
Exit params	TP=10×ATR, SL=2.0×ATR, Trail=6.0×A
Risk / Lev	3.0% per trade, 3.0× cap
Trades (full)	409
Win rate	40.3%
Profit factor	1.44
Avg leverage	0.58×
Exposure	50.6%
CAGR (net)	**+37.4%**
Sharpe (annualized)	+1.09
Max drawdown	-38.4%
Final equity ($10k)	$111,621
IS (2019-2023)	n=280 CAGR +47.6% Sh +1.26 DD -3
OOS (2024-2026)	n=129 CAGR +17.7% Sh +0.70 DD -3
Walk-forward verdict	✓ **OOS holds**

Walk-forward OOS (split 2024-01-01): IS Sharpe +1.26 / OOS Sharpe +0.70 (+17.7% OOS CAGR, n=129). Verdict: ✓ **OOS holds**.

Notes. Signal parameters above are native-TF values (already scaled from 1h). This configuration was selected by grid search over the 2019-2026 full window. Equity curve is on a log scale. Drawdown is the peak-to-trough equity % from the prior high-water mark. Monthly-returns heatmap colors: green ≥ 0, red < 0, clamped at ±30% for visibility.

AVAX — RangeKalman_LS @ 4h — CAGR +77.5%, Sharpe +1.49, DD -31.1%

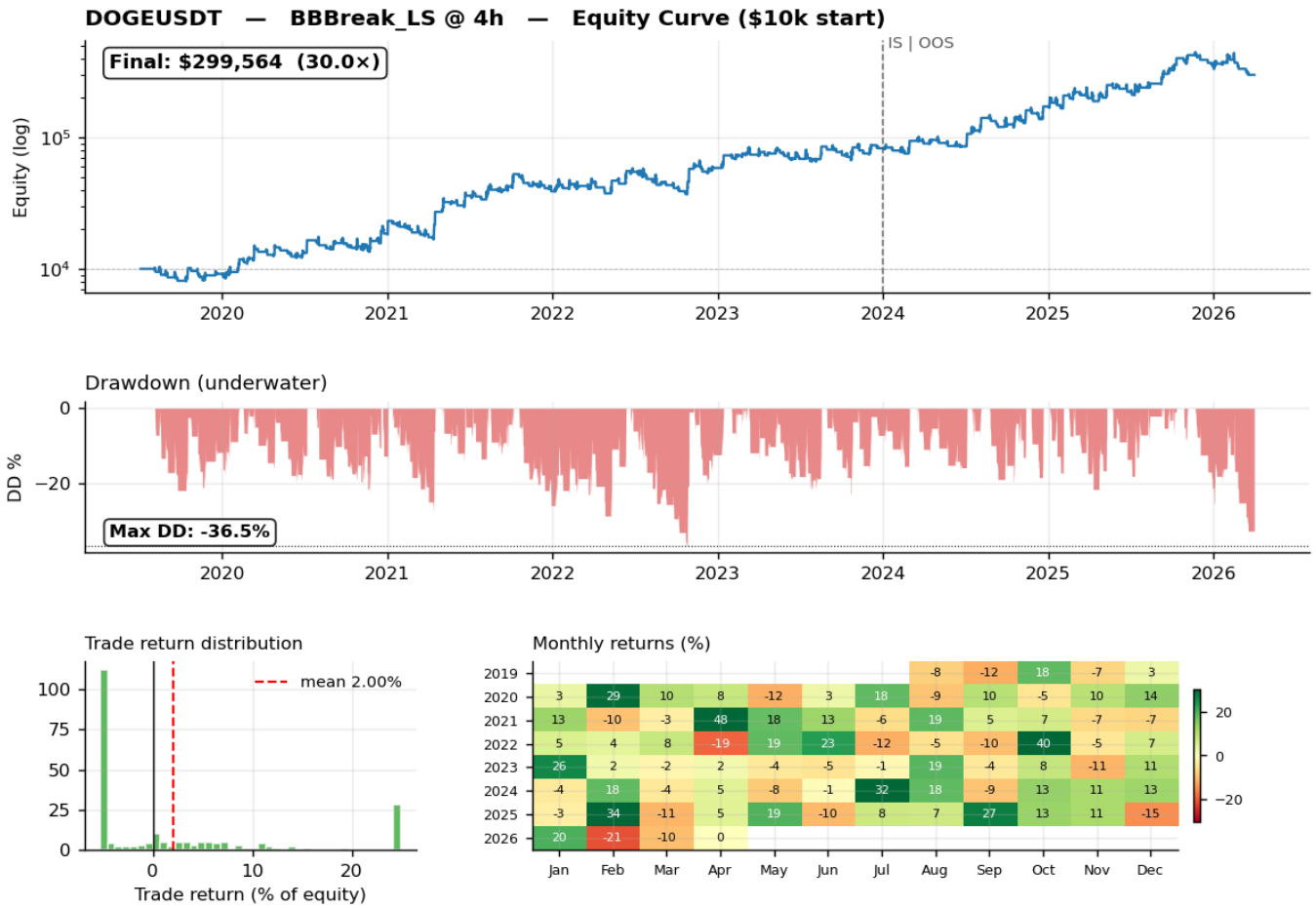


Strategy family	RangeKalman_LS
Timeframe	4h
Signal params	alpha=0.07, rng_len=50, rng_mult=2.5,
Exit params	TP=10×ATR, SL=2.0×ATR, Trail=6.0×A
Risk / Lev	5.0% per trade, 3.0× cap
Trades (full)	165
Win rate	49.7%
Profit factor	2.01
Avg leverage	0.94×
Exposure	27.7%
CAGR (net)	**+77.5%**
Sharpe (annualized)	+1.49
Max drawdown	-31.1%
Final equity ($10k)	$253,687
IS (2019-2023)	n=100 CAGR +95.9% Sh +1.68 DD -3
OOS (2024-2026)	n=65 CAGR +53.7% Sh +1.19 DD -30
Walk-forward verdict	**✓ OOS holds**

Walk-forward OOS (split 2024-01-01): IS Sharpe +1.68 / OOS Sharpe +1.19 (+53.7% OOS CAGR, n=65). Verdict: **✓ OOS holds**.

Notes. Signal parameters above are native-TF values (already scaled from 1h). This configuration was selected by grid search over the 2019-2026 full window. Equity curve is on a log scale. Drawdown is the peak-to-trough equity % from the prior high-water mark. Monthly-returns heatmap colors: green ≥ 0, red < 0, clamped at ±30% for visibility.

DOGE — BBBreak_LS @ 4h — CAGR +63.5%, Sharpe +1.22, DD -36.5%

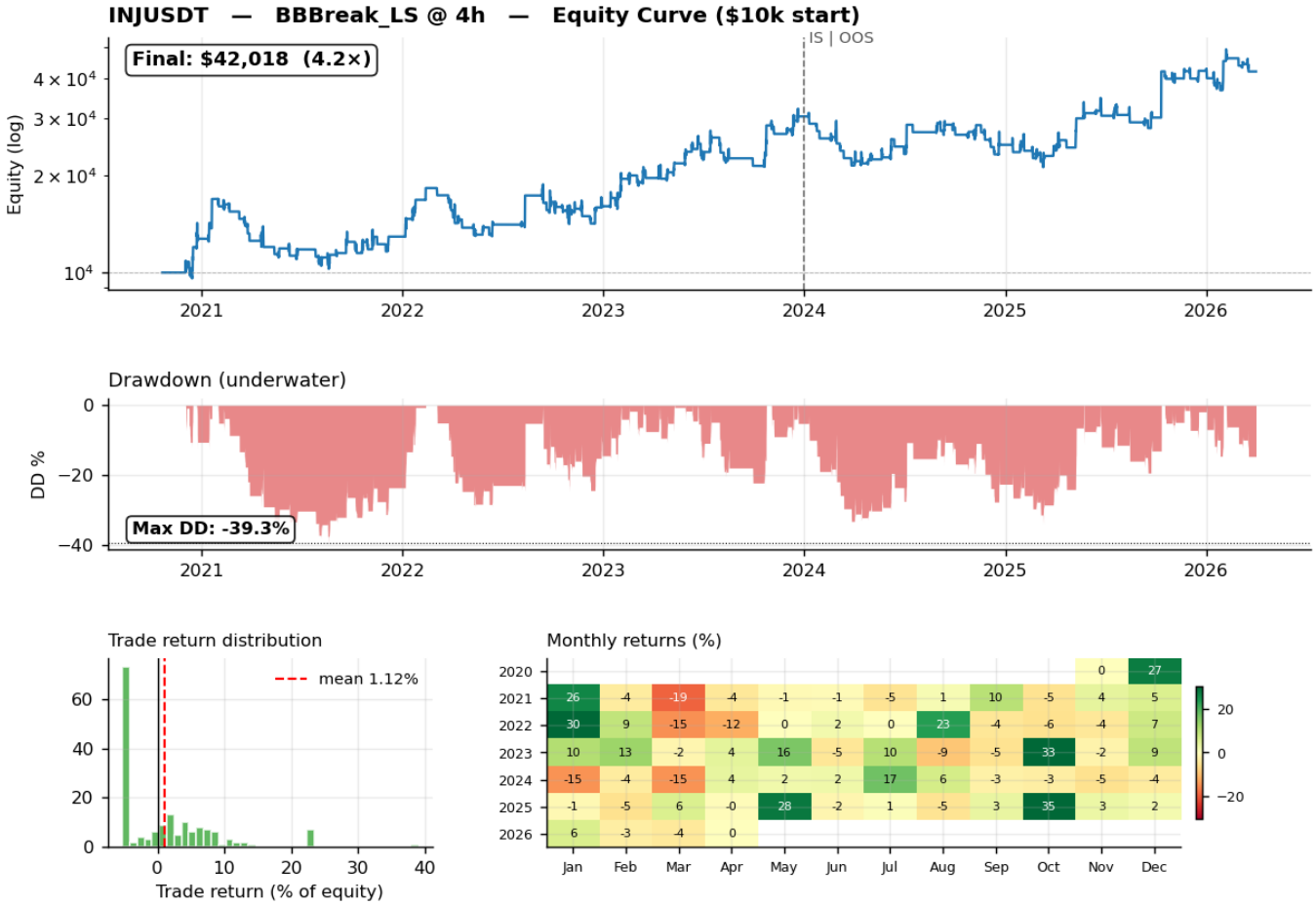


Strategy family	BBBreak_LS
Timeframe	4h
Signal params	n=45, k=2.5, regime_len=75
Exit params	TP=10×ATR, SL=2.0×ATR, Trail=6.0×A
Risk / Lev	5.0% per trade, 3.0× cap
Trades (full)	222
Win rate	41.9%
Profit factor	1.73
Avg leverage	1.03×
Exposure	25.3%
CAGR (net)	**+63.5%**
Sharpe (annualized)	+1.22
Max drawdown	-36.5%
Final equity ($10k)	$299,564
IS (2019-2023)	n=151 CAGR +57.8% Sh +1.14 DD -3
OOS (2024-2026)	n=71 CAGR +75.0% Sh +1.39 DD -32
Walk-forward verdict	**✓ OOS holds**

Walk-forward OOS (split 2024-01-01): IS Sharpe +1.14 / OOS Sharpe +1.39 (+75.0% OOS CAGR, n=71). Verdict: **✓ OOS holds**.

Notes. Signal parameters above are native-TF values (already scaled from 1h). This configuration was selected by grid search over the 2019-2026 full window. Equity curve is on a log scale. Drawdown is the peak-to-trough equity % from the prior high-water mark. Monthly-returns heatmap colors: green ≥ 0, red < 0, clamped at ±30% for visibility.

INJ — BBBreak_LS @ 4h — CAGR +29.3%, Sharpe +0.79, DD -39.3%

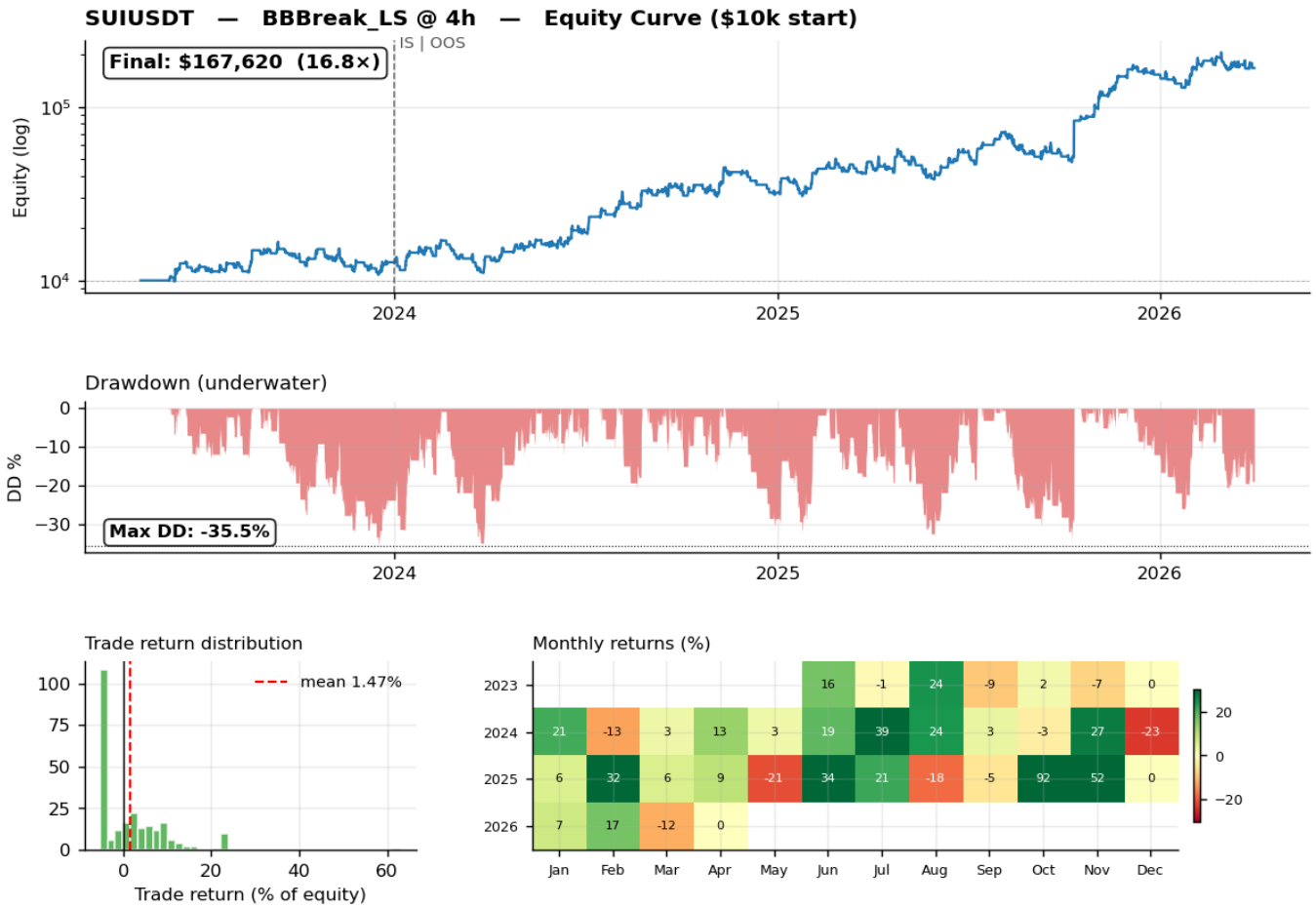


Strategy family	BBBreak_LS
Timeframe	4h
Signal params	n=30, k=2.5, regime_len=225
Exit params	TP=7×ATR, SL=1.5×ATR, Trail=4.5×AT
Risk / Lev	5.0% per trade, 3.0× cap
Trades (full)	169
Win rate	48.5%
Profit factor	1.48
Avg leverage	0.96×
Exposure	11.8%
CAGR (net)	**+29.3%**
Sharpe (annualized)	+0.79
Max drawdown	-39.3%
Final equity ($10k)	$42,018
IS (2019-2023)	n=101 CAGR +40.9% Sh +0.97 DD -3
OOS (2024-2026)	n=68 CAGR +14.4% Sh +0.52 DD -31
Walk-forward verdict	**✓ OOS holds**

Walk-forward OOS (split 2024-01-01): IS Sharpe +0.97 / OOS Sharpe +0.52 (+14.4% OOS CAGR, n=68). Verdict: **✓ OOS holds**.

Notes. Signal parameters above are native-TF values (already scaled from 1h). This configuration was selected by grid search over the 2019-2026 full window. Equity curve is on a log scale. Drawdown is the peak-to-trough equity % from the prior high-water mark. Monthly-returns heatmap colors: green ≥ 0, red < 0, clamped at ±30% for visibility.

SUI — BBBreak_LS @ 4h — CAGR +160.4%, Sharpe +1.66, DD -35.5%

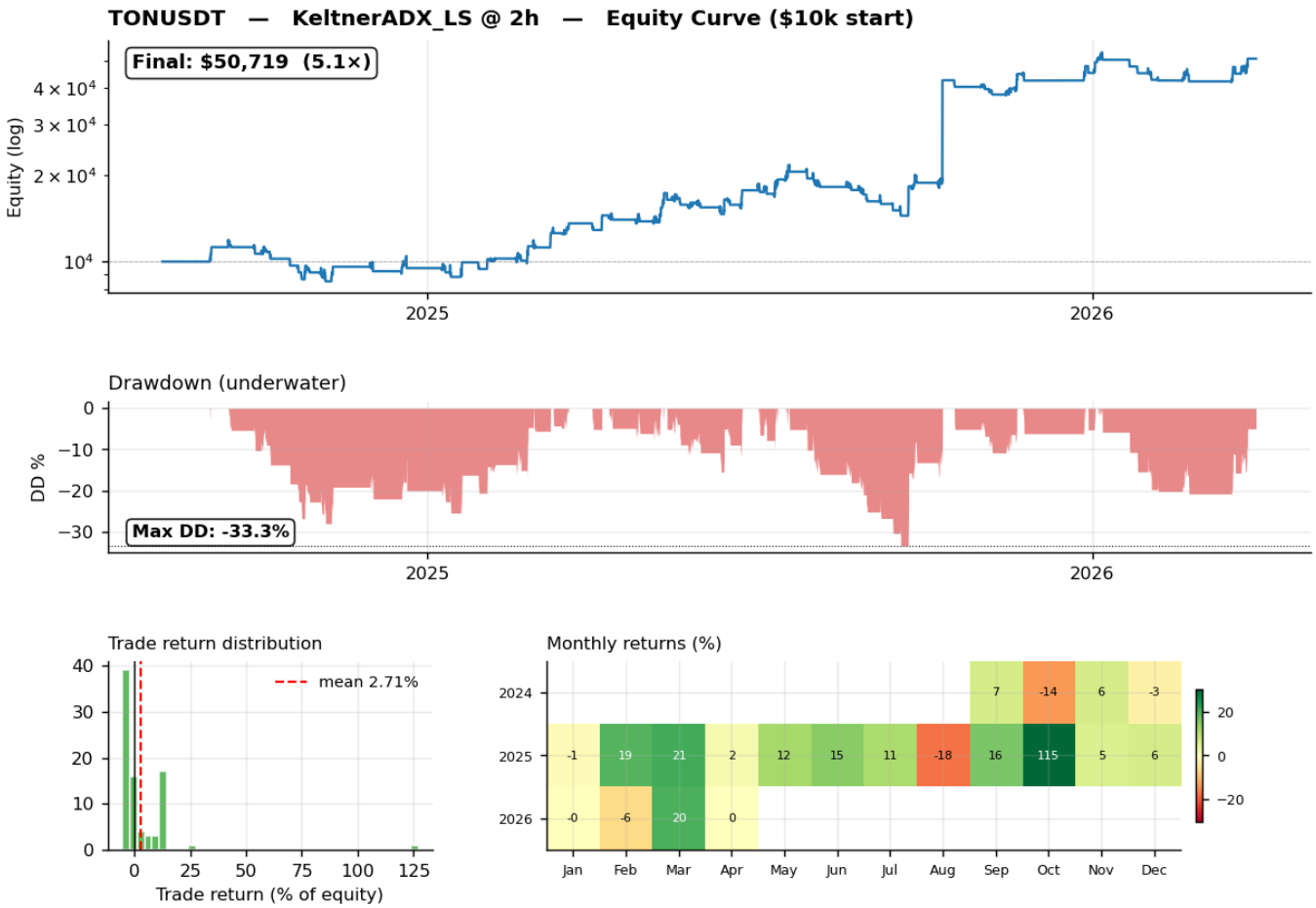


Strategy family	BBBreak_LS
Timeframe	4h
Signal params	n=15, k=1.5, regime_len=150
Exit params	TP=7×ATR, SL=1.5×ATR, Trail=4.5×AT
Risk / Lev	5.0% per trade, 3.0× cap
Trades (full)	245
Win rate	47.3%
Profit factor	1.62
Avg leverage	1.12×
Exposure	32.5%
CAGR (net)	**+160.4%**
Sharpe (annualized)	+1.66
Max drawdown	-35.5%
Final equity ($10k)	$167,620
IS (2019-2023)	n=50 CAGR +41.0% Sh +0.89 DD -35
OOS (2024-2026)	n=195 CAGR +212.0% Sh +1.85 DD -
Walk-forward verdict	**✓ OOS holds**

Walk-forward OOS (split 2024-01-01): IS Sharpe +0.89 / OOS Sharpe +1.85 (+212.0% OOS CAGR, n=195). Verdict: **✓ OOS holds**.

Notes. Signal parameters above are native-TF values (already scaled from 1h). This configuration was selected by grid search over the 2019-2026 full window. Equity curve is on a log scale. Drawdown is the peak-to-trough equity % from the prior high-water mark. Monthly-returns heatmap colors: green ≥ 0, red < 0, clamped at ±30% for visibility.

TON — KeltnerADX_LS @ 2h — CAGR +166.2%, Sharpe +1.31, DD -33.3%



Strategy family	KeltnerADX_LS
Timeframe	2h
Signal params	k_n=30, k_mult=2.0, adx_min=22, regin
Exit params	TP=5×ATR, SL=2.0×ATR, Trail=3.5×AT
Risk / Lev	5.0% per trade, 3.0× cap
Trades (full)	84
Win rate	40.5%
Profit factor	2.24
Avg leverage	1.66×
Exposure	16.5%
CAGR (net)	**+166.2%**
Sharpe (annualized)	+1.31
Max drawdown	-33.3%
Final equity ($10k)	$50,719
IS (2019-2023)	n=0 n/a n/a n/a
OOS (2024-2026)	n=84 CAGR +166.2% Sh +1.31 DD -3
Walk-forward verdict	OOS-only history (no IS)

Walk-forward OOS: no IS slice (post-2024 listing) — full-sample IS Sharpe +1.31, OOS-only history (no IS).

Notes. Signal parameters above are native-TF values (already scaled from 1h). This configuration was selected by grid search over the 2019-2026 full window. Equity curve is on a log scale. Drawdown is the peak-to-trough equity % from the prior high-water mark. Monthly-returns heatmap colors: green ≥ 0, red < 0, clamped at ±30% for visibility.

Methodology and caveats

Data. Binance perpetual-futures USDT-margined OHLCV bars. BTC/ETH/SOL histories start 2019, altcoins start when they first listed on Binance (SUI 2023-05, TON 2022-10). Loaded from the local features/multi_tf/ parquet store at 1h, 2h, and 4h.

Signal families tested (all long + short). (1) RangeKalman L+S — mean-reverting Kalman-EMA with deviation bands, direction gated by a regime SMA. (2) BB-Break L+S — Bollinger band breakout + regime SMA gate. (3) Keltner + ADX L+S — Keltner channel breakout gated by ADX strength + regime SMA. For each coin, the best config was chosen from a small per-family grid (α, lengths, multipliers, risk, TF).

Execution model. Signals fire at bar close; fills at next-bar open ± 3 bps slippage. Exits: TP at +tp×ATR, SL at -sl×ATR, trailing stop at -trail×ATR from high-water mark, forced exit at max-hold bars. Position sizing: risk_per_trade × equity ÷ (sl×ATR), capped at leverage_cap × equity ÷ price. Taker fee of 0.045% charged on entry and exit notional.

Performance accounting. CAGR is compounded annual return on equity from a $10,000 start. "Net" CAGR subtracts a modeled 8% APR funding drag × average leverage × time-in-market. Sharpe is annualized using the natural bar count per year at each TF. Drawdown is the maximum peak-to-trough decline of the equity curve. Profit factor is sum(winning P&L;) / |sum(losing P&L;)|.

What this report IS. A per-coin optimization showing the best single-config for each asset, with a walk-forward out-of-sample check. For each coin the same locked config is re-simulated in two slices: IS (2019-2023) and OOS (2024-2026). The "Walk-forward verdict" column flags a config as holding when OOS Sharpe ≥ 50% of IS Sharpe. TON has no IS history (first listing 2024-08) and is shown OOS-only.

What this report is NOT. A go-live spec. Expect 20-40% haircut on CAGR from slippage, borrowing frictions, venue outages, and regime change. Paper trade each coin for 4+ weeks before real capital; set a -45% circuit breaker per sub-account; re-audit every 6 months. LINK (37%) and INJ (29%) fall short of the 55% target — use them for diversification, not as standalone aggressive allocations.